

RIDA MELKAOU

INDUSTRIAL ENGINEER | DATA, AI & BUSINESS INTELLIGENCE

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LinkedIn | GitHub



PROFILE

Industrial and Production Engineer combining supplier-quality ownership with data analytics, BI and applied AI. Builds traceable decision systems with Python, SQL, Power BI and statistical/optimization methods - from launch and issue visibility to demand planning, routing and AI-agent reliability.

PROFESSIONAL EXPERIENCE

MAGNA INTERNATIONAL

Supplier Quality Engineer - Electronics & Metal Commodities

Kenitra, Morocco

Mar 2026 - Present

- Manage supplier quality across Europe, Asia and Morocco; use performance indicators, launch readiness, 8D/CAPA status and open-issue risk to prioritize action.
- Developed an SQE BI system covering nomination through Safe Launch and SOP, consolidating activities, critical-part status, milestones and open issues into one traceable management view.
- Built a smart notification pipeline for overdue actions and launch risks, keeping owners and deadlines visible across the project portfolio.

STELLANTIS R&D

Engineering Intern - Data Analytics, Process Optimization & AI

Casablanca, Morocco

Feb 2025 - Aug 2025

- Mapped a configuration-anomaly workflow and built a Python application with a structured Databricks data model for detection, prioritization and traceability; reduced treatment time from 2 days to 2 minutes.

SANY - FORGE DE BAZAS

Engineering Intern - Process Digitalization & Maintenance

Casablanca, Morocco

Jul 2024 - Sep 2024

- Built a Unity/NLP assistant to structure access to maintenance and spare-parts knowledge, reducing fragmented search and improving shop-floor information flow.

3D SMART FACTORY

Data Analyst Intern

Mohammedia, Morocco

Aug 2023 - Sep 2023

- Built Power BI and Excel reporting around OEE, FPY, scrap and downtime to expose production bottlenecks for Lean review.

SELECTED DECISION SYSTEMS

Demand / Order | Forecasting and replenishment

Built a cutoff-safe Python decision engine across 900 M5 item-store series; improved WAPE 4.5% vs the strongest calibration-selected seasonal baseline and simulated 1,528 fewer lost units at 96.7% fulfilled demand. [Evidence](#)

Route / Control | Optimization and operations

Learned zone sequencing from 6,112 historical routes and built a constrained route policy; on a separate 13-route public cohort, cut simulated drive time 2.8% vs nearest neighbour, maintained 100% timed-package adherence and eliminated 332 zone re-entries.

[Evidence](#)

Agent / Proof | AI evaluation and observability

Built a reliability lab over 3,336 official historical tool-agent trajectories; implemented Pass1-Pass4, cost, latency and failure diagnostics, exposing a 14.9-point one-shot-to-repeatability loss in the pinned GPT-4.1 telecom cohort. [Evidence](#)

EDUCATION

Mohammed VI International Civil Aviation Academy (AIAC)

Nouaceur, Morocco

2022 - 2025

Engineering Degree - Industrial & Production Engineering

Salmene El Farissi Preparatory Classes

Sale, Morocco

MPSI - MP

2020 - 2022

SKILLS

Data & AI: Python, SQL, Pandas, NumPy, scikit-learn, statistics, forecasting, experimentation, data modeling, data quality, Databricks, Git, Docker.

BI & Delivery: Power BI, DAX, Excel, Power Query, Tableau, React, TypeScript, FastAPI, dashboard design, KPI systems, automation, reporting.

Industrial & Quality: supplier/process performance, PPM, OEE, FPY, scrap, downtime, 8D/CAPA, FMEA, root-cause analysis, Lean, launch/SOP governance.

LANGUAGES & CERTIFICATIONS

Languages: Arabic - Native | French - Fluent | English - C1 (TOEIC 855) Certifications: [IBM Data Analyst Professional Certificate \(2026\)](#); [EF SET C1](#); [TOEIC 855](#)